

Normalization Lab

1. DreamHome

Company designed to facilitate connecting people who are interested in properties renting and the owners of these properties. Note that 'Rent' is the value of the property cost, which determined by the negotiation between the customer and the owner. Use the below data to reach the 3rd Normal Form. Show your answer steps. Start with Customer number as a PK. Ignore Page number and Date mentioned in the below bill.

Page 1		DreamHome Customer Rental Details				Date 7-Oct-98	
Customer Name John Kay				Customer Number CR76			
Property Number	Property Address	Rent Start	Rent Finish	Rent	Owner Number	Owner Name	
PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy	
PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw	

Customer_Rental Table								
Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
		PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-June-94	350	CO40	Tina Murphy
		PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
		PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

Step 1: 1st Normal Form (1NF)

Rule: Eliminate repeating groups → one row per rental instance

Customer_Rental_1NF

Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR76	John Kay	PG4	6 Lawrence St, Glasgow	1-Jul-94	31-Aug-96	350	CO40	Tina Murphy
CR76	John Kay	PG16	5 Novar Dr, Glasgow	1-Sep-96	1-Sep-98	450	CO93	Tony Shaw

Customer_No	CName	Property_No	PAddress	RentStart	RentFinish	Rent	Owner_No	OName
CR56	Aline Stewart	PG4	6 Lawrence St, Glasgow	1-Sep-92	10-Jun-94	350	CO40	Tina Murphy
CR56	Aline Stewart	PG36	2 Manor Rd, Glasgow	10-Oct-94	1-Dec-95	375	CO93	Tony Shaw
CR56	Aline Stewart	PG16	5 Novar Dr, Glasgow	1-Jan-96	10-Aug-96	450	CO93	Tony Shaw

1NF Achieved: Atomic values, no repeating groups

Step 2: 2nd Normal Form (2NF)

Rule: No Partial Dependency

2NF Tables

- Cutomer Table

Customer_No (PK)	CName
CR76	John Kay
CR56	Aline Stewart

- Owner Table

Owner_No (PK)	OName
CO40	Tina Murphy
CO93	Tony Shaw

- Proberity Table

Property_No (PK)	PAddress	Owner_No (FK)
PG4	6 Lawrence St, Glasgow	CO40
PG16	5 Novar Dr, Glasgow	CO93

- Rental Table

Customer_No (FK)	Property_No (FK)	RentStart (PK)	RentFinish	Rent
CR76	PG4	1-Jul-94	31-Aug-96	350
CR76	PG16	1-Sep-96	1-Sep-98	450

Step 3: 3nd Normal Form (3NF)

Rule: No Transitive Dependency

3NF Tables

- Cutomer Table

PK: Customer_No

Customer_No (PK)	CName
CR76	John Kay
CR56	Aline Stewart

• Owner Table

PK: Owner_No

Owner_No (PK)	OName
CO40	Tina Murphy
CO93	Tony Shaw

• Proberity Table

PK: Property_No

Property_No (PK)	PAddress	Owner_No (FK)
PG4	6 Lawrence St, Glasgow	CO40
PG16	5 Novar Dr, Glasgow	CO93

• Rental Table

PK: RentStart

Customer_No (FK)	Property_No (FK)	RentStart (PK)	RentFinish	Rent
CR76	PG4	1-Jul-94	31-Aug-96	350
CR76	PG16	1-Sep-96	1-Sep-98	450

Sample Report Layout

The below report shows detailed information about the organization projects and the employees work for. As shown in below records; the project may have many employees; also, an employee may work for more than one project. Each job classification has a specific hourly rate (CHG/Hour). You are required to apply the 1st, 2nd and 3rd NF.

TABLE 5.1 A Sample Report Layout							
PROJ. NUM.	PROJECT NAME	EMPLOYEE NUMBER	EMPLOYEE NAME	JOB CLASS.	CHG/HOUR	HOURS BILLED	TOTAL CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	\$ 85.50	23.8	\$ 2,011.10
		101	John G. News	Database Designer	\$105.00	19.4	\$ 2,037.00
		105	Alice K. Johnson*	Database Designer	\$105.00	35.7	\$ 3,748.50
		106	William Smithfield	Programmer	\$ 35.75	12.6	\$ 450.45
		102	David H. Senior	Systems Analyst	\$ 96.75	23.8	\$ 2,302.65
				Subtotal			\$10,549.70
18	Amber Wave	114	Annelise Jones	Applications Designer	\$ 48.10	25.6	\$ 1,183.26
		118	James J. Frommer	General Support	\$ 18.36	45.3	\$ 831.71
		104	Anne K. Ramoras*	Systems Analyst	\$ 96.75	32.4	\$ 3,135.70
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	45.0	\$ 2,021.00
				Subtotal			\$ 7,172.47
22	Rolling Tide	105	Alice K. Johnson	Database Designer	\$105.00	65.7	\$ 6,793.50
		104	Anne K. Ramoras	Systems Analyst	\$ 96.75	48.4	\$ 4,682.70
		113	Delbert K. Joenbrood*	Applications Designer	\$ 48.10	23.6	\$ 1,135.16
		111	Geoff B. Wabash	Clerical Support	\$ 26.87	22.0	\$ 591.14
		106	William Smithfield	Programmer	\$ 35.75	12.8	\$ 457.60
				Subtotal			\$13,660.10
25	Starlight	107	Maria D. Alonzo	Programmer	\$ 35.75	25.6	\$ 879.45
		115	Travis B. Bawangi	Systems Analyst	\$ 96.75	45.8	\$ 4,431.15
		101	John G. News*	Database Designer	\$105.00	56.3	\$ 5,911.50
		114	Annelise Jones	Applications Designer	\$ 48.10	33.1	\$ 1,592.11
		108	Ralph B. Washington	Systems Analyst	\$ 96.75	23.6	\$ 2,283.30
		118	James J. Frommer	General Support	\$ 18.36	30.5	\$ 559.98
		112	Darlene M. Smithson	DSS Analyst	\$ 45.95	41.4	\$ 1,902.33
				Subtotal			\$17,559.02
				Total			\$48,942.09
Note: * indicates project leader.							

1NF

- Elemenating Repeting groups, only atomic information

PROJ_NUM	PROJECT NAME	EMP_NUM	EMP_NAME	JOB_CLASS	CHG_HOUR	HOURS_BILLED	TOTAL_CHARGE
15	Evergreen	103	June E. Arbough	Elec. Engineer	85.50	23.8	2031.10
15	Evergreen	101	John G. News	Database Designer	105.00	19.4	2037.00
15	Evergreen	105	Alice K. Johnson	Database Designer	105.00	35.7	3748.50
15	Evergreen	106	William Smithfield	Programmer	35.75	12.6	450.45
15	Evergreen	102	David H. Senior	Systems Analyst	96.75	23.8	2302.65

2NF

- Rule: No Partial Dependency

Employee Table

EMP_NUM (PK)	EMP_NAME	JOB_CLASS	CHG_HOUR
101	John G. News	Database Designer	105.00
102	David H. Senior	Systems Analyst	96.75
103	June E. Arbough	Elec. Engineer	85.50
104	Anne K. Ramoras	Systems Analyst	96.75
105	Alice K. Johnson	Database Designer	105.00

Project Table

PROJ_NUM (PK)	PROJECT_NAME
15	Evergreen
18	Amber Wave
22	Rolling Tide
25	Starlight

Assigned Table

PROJ_NUM (FK)	EMP_NUM (FK)	HOURS_BILLED
15	103	23.8
15	101	19.4
15	105	35.7
15	106	12.6
15	102	23.8

3NF

- **Rule:** No Transative Dependency

Employee Table

PK: EMP_NUM

EMP_NUM (PK)	EMP_NAME
101	John G. News
102	David H. Senior
103	June E. Arbough
104	Anne K. Ramoras
105	Alice K. Johnson

Job_class Table

PK: JOB_CLASS

JOB_CLASS (PK)	CHG_HOUR
Database Designer	105.00
Systems Analyst	96.75
Elec. Engineer	85.50
Systems Analyst	96.75
Database Designer	105.00

Project Table

PK: PROJ_NUM

PROJ_NUM (PK)	PROJECT_NAME
15	Evergreen
18	Amber Wave
22	Rolling Tide
25	Starlight

Assigned Table

PK: (PROJ_NUM, EMP_NUM)
FK: PROJ_NUM -> Project
FK: EMP_NUM -> Employee

PROJ_NUM (FK)	EMP_NUM (FK)	HOURS_BILLED
15	103	23.8
15	101	19.4
15	105	35.7
15	106	12.6
15	102	23.8